

Q9. Chart Selection Justification

Different chart types were used in this analysis based on the nature of each business question and the type of insight required.

1. Bar Charts (Category & Region Analysis)

Bar charts were used to compare **total Sales across Categories and Regions** because they make it easy to:

- Compare values across multiple groups
- Rank performance (highest to lowest)
- Clearly identify top-performing categories or regions

👉 Best for: **ranking and comparison**

2. Line Chart (Sales Trend Over Time)

A line chart was used for **monthly sales trends** because it effectively shows:

- Changes over time
- Growth patterns
- Seasonal fluctuations and trends

👉 Best for: **time-series analysis**

3. Scatter Plot (Sales vs Profit)

A scatter plot was used to analyze the relationship between **Sales and Profit** because it:

- Shows correlation between two continuous variables
- Helps identify outliers (high sales but low profit)
- Highlights loss-making orders clearly

👉 Best for: **relationship and anomaly detection**

4. Heatmap (Category vs Region)

A heatmap was used to compare **Category performance across Regions** because it:

- Displays two-dimensional comparisons clearly
- Highlights strong and weak areas using color intensity
- Makes patterns easy to interpret visually

👉 Best for: **matrix-based comparison**

5. Bubble Chart (Product Prioritization)

A bubble chart was used to analyze **Sales, Profit, and Quantity together** because it:

- Combines multiple variables in one view
- Shows magnitude (size = quantity)
- Shows impact (position = sales & profit)

👉 Best for: **multi-dimensional analysis**

Q10. Business Insight Summary

Insight 1: Certain Categories and Regions Drive Most Revenue

The analysis shows that only a few **categories and regions contribute the majority of total sales**, while others perform significantly lower.

Business Impact:

- Management can focus marketing and sales efforts on high-performing regions/categories
 - Underperforming regions can be targeted with promotions or operational improvements
 - Helps allocate resources more efficiently
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Insight 2: High Sales Does Not Always Mean High Profit

The Sales vs Profit analysis reveals that some **high-sales orders or products generate low or even negative profit**.

Business Impact:

- Indicates possible issues like heavy discounting or high operational costs
 - Helps finance and sales teams review pricing strategies
 - Enables the company to focus on **profitable growth instead of just revenue growth**
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Overall Value to Management

These insights help leadership:

- Make data-driven strategic decisions
- Improve profitability, not just sales volume
- Identify growth opportunities and inefficiencies

- Optimize regional and product-level performance